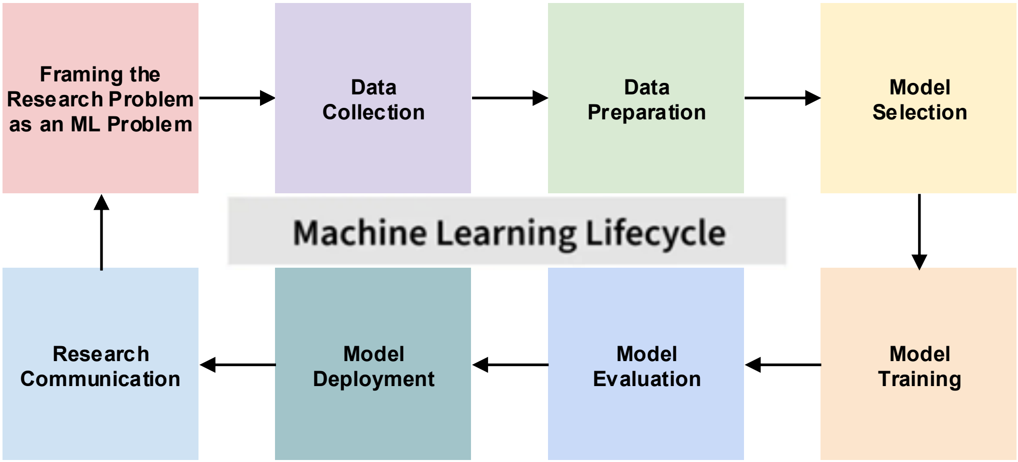
AI/ML Lifecycle

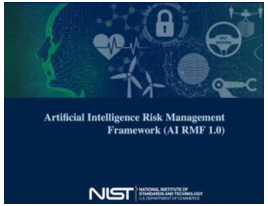


Good AI isn't just accurate
it is also responsible, trustworthy, and appropriate for its intended use.

Responsible AI

The practice of designing, developing, deploying, and using AI systems in ways that identify, evaluate, and reduce potential risks and harms while maximizing benefits throughout the AI lifecycle.

- Who/What could be affected/harmed?
- What could go wrong or cause harm?
- How likely is that harm, and how would we know?
- What can we do to reduce that risk?



Source: nist.gov

EU Artificial Intelligence Act

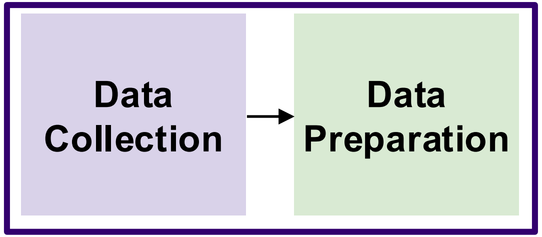
Source: artificialintelligenceact.eu

NIST Artificial intelligence risk management framework (AI RMF 1.0)
The EU Artificial Intelligence Act

Framing the Research Problem as an ML Problem

Should AI even be used for this problem?

- What decision is your AI supporting?
- Is AI making the decision, assisting a human decision, learning unseen patterns, or automating a repetitive task?



Data Governance

Privacy/Confidentiality/Security/Trust

- Where is the data coming from? Where is it stored?
- Does it contain sensitive or personally identifiable information?
- Do we have permission to use the data?

Representation / Bias

- Is it representative of the population or scenarios?

Data Quality

- What assumptions are you introducing while cleaning or labeling the data?

Model
Selection

Explainability or Transparency

- Which model is appropriate for this problem?
- Have you chosen the simplest model that adequately solves your problem?
- Could explainability or transparency matter more than a small gain in accuracy?

Model Training

+ Robustness

Privacy

- Is model memorizing sensitive information?

Representation / Bias

- Is it introducing additional biases?

Robustness

- With change in data or environment (distribution shifts), how confident are you that it will still behave appropriately?

Model Evaluation

Holistic Evaluation and Auditing

How do we know our model is actually good?

- What metrics tell whether system is trustworthy?
    - Does it perform equally well across different subgroups, or only well on average?
    - Have you tested it under conditions it wasn't trained on: distribution shift, edge cases?
- Auditing: Has it been reviewed by someone other than the person who built it?

Model Deployment

Accountability & Oversight

Monitoring

- Is anyone tracking performance after launch, as real-world conditions shift?

Human Oversight

- Is there an opportunity for a human to review, override, or appeal decisions?

Responsibility

- Who is accountable if the system makes a mistake?

Research Communication

Responsible Communication

- Are we communicating our findings honestly and transparently?
- If someone reads your paper or uses your tool without talking to you, will they understand its limitations and appropriate use?
- Are your claims proportionate to your evidence?